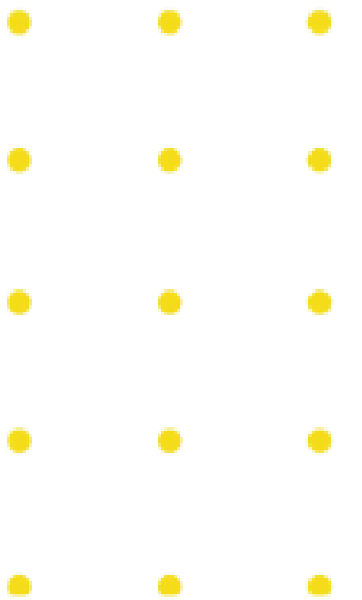
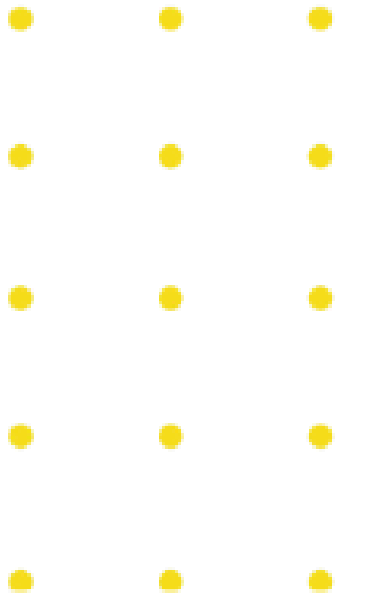


Module 5:

Networking Devices



Topic 2: MAC Addresses/IP Addresses

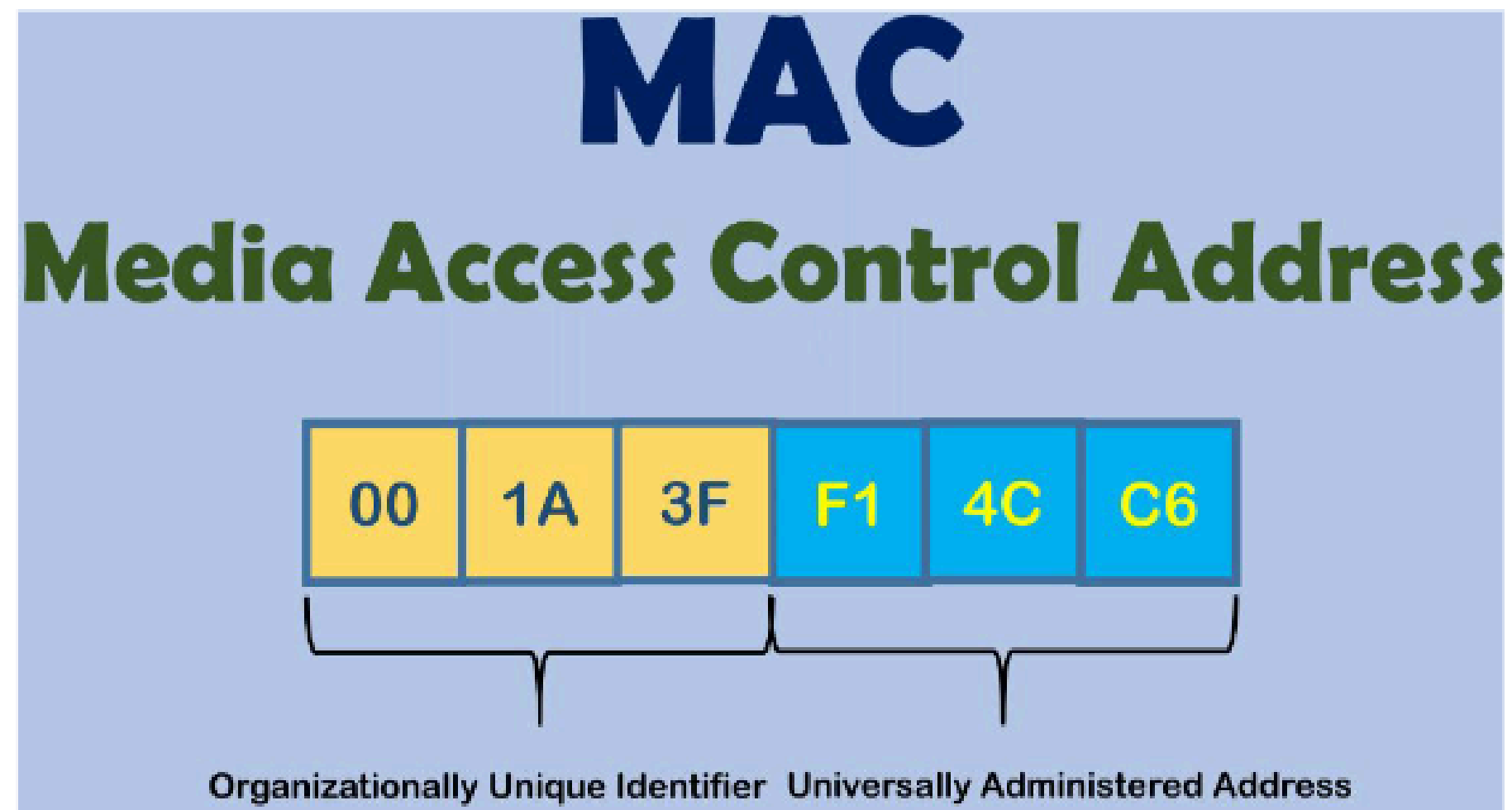


MAC Address

- The “MAC” part stands for “Media Access Control” and “address” part refers to a unique value (number).
- MAC address is a hardware identifier that **uniquely identifies each device on a network**. Primarily, the manufacturer assigns it.
- Each device you have (Mobile, PC, Laptop) that could connect the internet, has a **unique MAC Address**.
- In network level, devices are communication with their MAC Addresses using **Switches**.

MAC Address

- The *MAC address* is a 12 digit **hexadecimals**
 - MAC Address Example: **00:00:5e:00:53:af**



IP Address

- The “IP” stands for “Internet Protocol” and “address” part refers to a unique value (number).
- IP address is an assigned number linked to any device connected to the internet that identifies uniquely devices to enable communication.
- Data sent among devices must have **source IP Address** and **destination IP address** to have the right path from source to destination.
- IP address typical format X.X.X.X; where X varies from 0-255
 - Example: 192.168.1.5

IPv4 Vs IPv6

IP version 4 and IP version 6 have the same functionality but differs in some features that **IPv6** takes lead in covering some **IPv4** weaknesses points.

IPv4	vs.	IPv6
Deployed 1981		Deployed 1998
32-bit IP address		128-bit IP address
4.3 billion addresses		7.9x10 ²⁸ addresses
Addresses must be reused and masked		Every device can have a unique address
Numeric dot-decimal notation		Alphanumeric hexadecimal notation
192.168.5.18		50b2:6400:0000:0000:6c3a:b17d:0000:10a9 (Simplified - 50b2:6400::6c3a:b17d:0:10a9)
DHCP or manual configuration		Supports autoconfiguration

IPConfig

How to know your computer IP/MAC address?

```
C:\Users\user>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-1TG41RS
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Ethernet Connection (7) I219-V
Physical Address. . . . . : 0C-9D-92-84-93-6A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::6c12:cb7f:beb4:f5ca%14(Preferred)
IPv4 Address. . . . . : 192.168.1.16(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, May 24, 2022 12:31:56 PM
Lease Expires . . . . . : Tuesday, May 24, 2022 4:04:54 PM
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 101490066
DHCPv6 Client DUID. . . . . : 00-01-00-01-28-E4-03-43-0C-9D-92-84-93-6A
DNS Servers . . . . . : 163.121.128.134
                       163.121.128.135
                       192.168.1.1
                       fe80::1%14
```

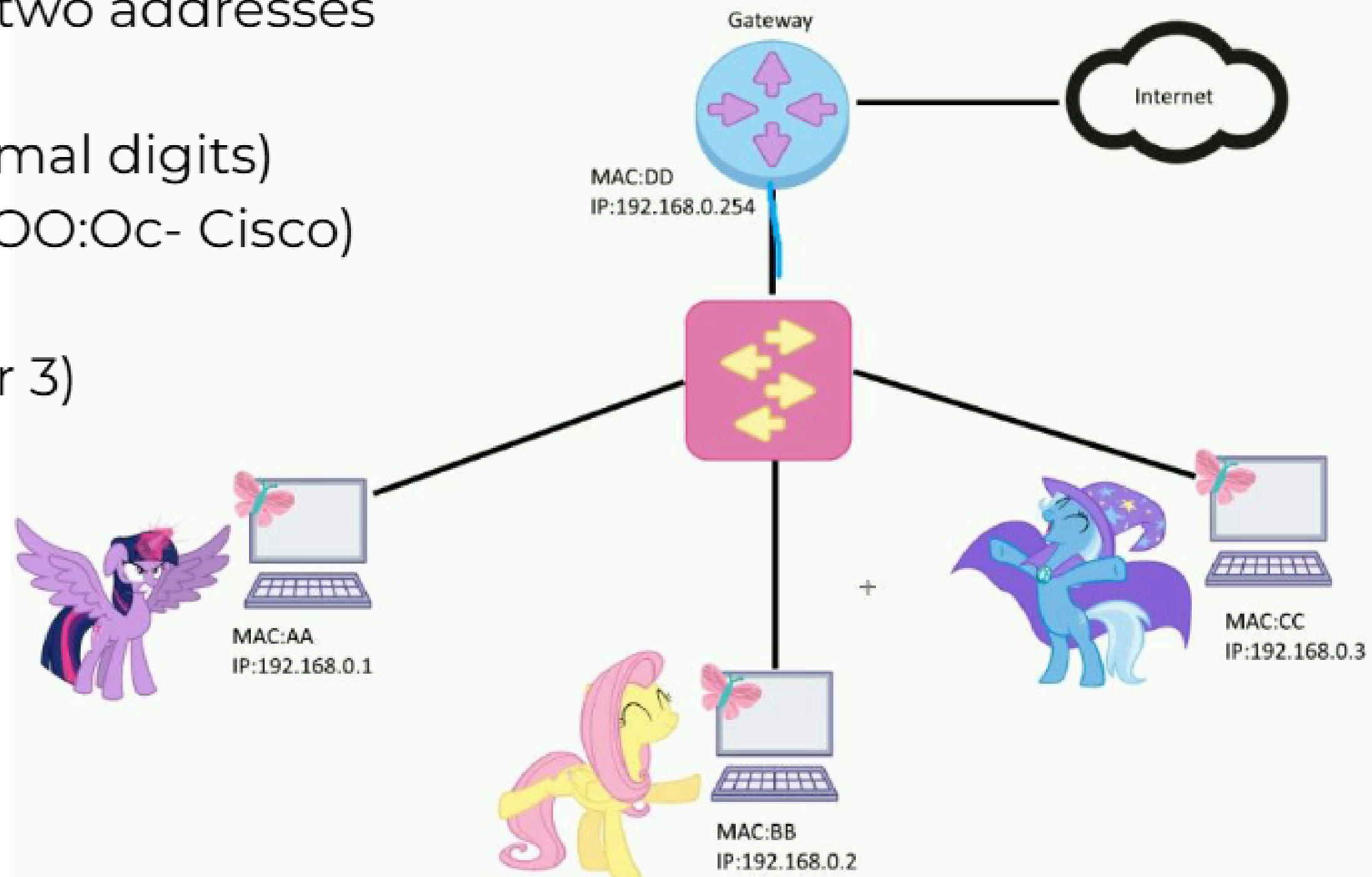
MAC Vs IP Addresses

At a minimum a computer has two addresses
MAC address (Layer 2)

- 48 bit address (12 hexadecimal digits)
- First half vendor code (00:00:0c- Cisco)

IP address is configurable (Layer 3)

- 32 bit address
- Part network and part host
- Configured by user



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